ZOO-501 Developmental Biology Mid term 2019

M.C.Qs

1. The birth defect study is called_____ ***teratology***
2. Zp3 function is_____ ***initiate acrosomal reaction***
3. There are ***three*** strategies of developmental commitment.
4. What is the regulative speciation_____
5. The human brain has _____ ganglia. ***21 or 22 pairs***
6. When potassium ions leaves the sperm then its membrane potential be __ ***Negative***
7. *Fertilin* proteins is important for sperm egg fusion in___ ***mammals***
8. Release of calcium from intracellular storage controlled by dye? ____ ***Aequorin***
9. Transition from fertilization to cleavage is caused by factors ___ ***MPF***
10. Eggs having sparse and equally yolk space are called _____ ***isolecithal***
11. Archenteron is a used for ____ ***Primitive gut***
12. Sonic hedgehog is type of ____***Protein***
13. Which type of Neurulation in fish ____ ***Secondary Neurulation***
14. Neural crest divides functionally in ***two*** domains.
15. Myogenic HLH proteins are specific for development of ___ ***Muscle Cell***
16. Identical twins occur in roughly ***0.25%*** of human births.
17. ***Regulative development*** is seen in most vertebrate embryos, and it is obviously critical in the development of *identical twins*.
18. Insects such as Drosophila use all ***three modes of specification*** to commit their cells to particular fates.
19. The speed of metamorphosis is ***directly*** linked with environmental pressures.
20. In ***acetabularia*** unicellular organism, "development" is controlled at both the ***transcriptional and translational levels***.
21. The individuals of each species, however, are divided into two mating types: ***plus and minus calamadomonas***
22. Cadherins. As their name suggests, they are ***calcium*** dependent adhesion molecules.
23. In some cases, the same transcription factors that activate the transcription of one gene can be used to repress the transcription of other genes. These "***negative enhancers*** " are also called *silencers*
24. The volume of a sea urchin egg is about ***200 picoliters*** (2 ×10-4 mm^3, more than 10,000 times the volume of the sperm).

25. Mutations of the ***SOX9 gene*** cause ***camptomelic dysplasia***.
26. The most crucial, difference between mammalian cleavage and all other types involves the phenomenon of ***compaction***.

27. The embryo contains ***three germ layers***: the ectoderm, the endoderm, and the mesoderm
28. The neural form ***keel fish***
29. Spiral holoblastic cleavage is characteristic of several animal groups, including annelid worms, some flatworms, and most mollusks ____ ***Snails***
30. Diploblastic animals ***Hydra*** are those who have ectoderm and endoderm, but no true mesoderm.
31. Cell fates are determined by signaling. The micromeres constitute a major signaling center. β-catenin is important for the inducing capacity of the micromeres
32. Even distribution of yolk in ovum ____ ***isolecith***
33. Yolk is present in ***center*** of centrolacithlans.
34. Eukaryotic genes are contained within a complex of DNA and protein called ____***Chromatin***
35. Regulation of gene expression can be accomplished by _____ ***Selective mRNA translation***
36. Red blood cells makes___ ***Globins***
37. The knowledge of gene activity in human can be obtained by ***Both A & B***
38. Transcription factors have ***three*** major domains
39. The technique which is in situ hybridization is ____ ***PCR***
40. A region of globular actin molecules lies between the nucleus and the acrosomal vesicle ***Sea Urchin***
41. A region that is not translated into protein_____ ***Untranslated region***
42. The mammalian egg is also surrounded by a layer of cells called ____ ***Cumulus***
43. Protein synthesis is accomplished by ribosomes and _____***tRNA***
44. Nucleosome is composed of an ***Octamer (8)*** of histone protein.
45. Development connects genotype and ____ ***Phenotype***
46. Major class of cadherin include _____***N-Cadherin***
47. Controlling of the efficiency and rate of transcription is ___***Enhancer***
48. Protocadherins play important in separating the notochord from the other mesodermal tissues during ***Xenopus*** gastrulation.
49. Which transmit genes from parents to offspring____***fertilization***
50. The RNA polymerase binds with ***promoter*** in the start of transcription.
51. Number of cortical granules Sea urchin____ ***15000***
52. Fertilin protein is essential for sperm-egg membrane fusion found in____ ***Mammals***
53. Which of the following uses all 3 modes of specification?____ ***Drosophila***
54. The 1st evidence suggesting the importance of sperm in reproduction shown by ***Lazzaro Spallanzani***
55. Nuleosomes contain approximately ***140*** base pairs of DNA.

56. Cadherins are ***Calcium*** ion dependent adhesion molecules.
57. Chordates are included in____ ***Deuterostomes***
58. In *Acetabularia* development is controlled at ***transcriptional and translational*** levels.
59. ***Estrogen*** is a hormone that can instruct which organ of the body to secrete the yolk proteins
60. Metamorphosis is initiated by hormones from the tadpole's____ ***Thyroid Gland***

61. J.R.Whitakar worked on blastomere to check out which of the following enzyme ***Acetylecholinesterase***
62. A condition in which cell lacks the instructions for development is____ ***Naïve***
63. Which of the following divide mitotically to produce all the cells of the body___ ***Zygote***
64. How many strategies involve in cell commitment___ ***2***
65. Which of the following Zona Pellucida of the mouse sperm binds __***ZP3***
66. The resting membrane potential of the egg is ____***70mv***
67. Calcium and bicarbonate ions may be critical in activating _____ ***cAMP***
68. One chemotactic molecule, a 14-amino acid peptide called ***Resact***
69. The force for sperm propulsion is provided by____ ***Dyenin***
70. Somites are produced by ____ ***Mesoderm***
71. Silencers are_____***Negative enhancers***
72. Brain has ____ ***100 Billion neurons***
73. Insects type of specification____ ***synctial specification***
74. ***Rana pipiens*** egg____ ***2500***
75. Multinucleated type of cell form____ ***syncytiotrophoblast***
76. Process of osteogensis____ ***2***
77. Spiral holoblastic cleavage in___ ***Snail***
78. ***Camptomelic dysplasia*** cause ___ ***sox9 mutation***
79. Human brain contains____ ***10^{12} glial Cells***
80. Telolecithal is cleavage in_____ ***Fishes***

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